

Solution Requirements

Date	08 October 2026
Team ID	UNM1611Firefox
Project Name	AI StudyBuddy API
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can register and securely log in using email/username and password. Passwords must be securely stored, and successful login provides authentication using JWT tokens.	High
2	Authorization levels	The system should control access to protected API resources so that only authenticated users can access and manage their own study materials and generated learning resources.	High
3	External interfaces	The AI StudyBuddy API connects with Google Gemini AI for generating summaries, flashcards, quizzes, and personalized study plans. The system also communicates with MongoDB for storing users, study materials, and	High

		services the system must connect to.	
4	Transactions processing	The system should process user registration, login, study-material upload, material retrieval, AI-generation requests, and storage/retrieval of generated learning resources through REST API endpoints.	High
5	Reporting	The system should provide generated learning outputs such as summaries, flashcards, quizzes, and study plans to the authenticated user in a structured format.	Medium
6	Business rules, etc.	Only authenticated users can access protected study materials and AI features. Uploaded material must be associated with the correct user. AI-generated content should be generated from the selected study material.	High
7	Compliance to laws or regulations	The system should protect user credentials and study materials and avoid exposing personal information or authentication tokens. API keys and sensitive configuration values must be stored securely using environment variables.	High
8	<i>Other</i>	The API should support study-material management and AI-powered learning features including summary generation, flashcard generation, quiz generation, and personalized study-plan generation.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	API requests should be processed efficiently, and normal CRUD operations should return responses quickly.	Normal API operations should respond within approximately 2 seconds under normal development conditions.
2	Scalability	The system should be designed so that additional users, study materials, and AI-generated resources can be supported as the application grows.	The backend should support increasing users and stored materials without requiring major changes to the core architecture.
3	Security & Data Privacy	User passwords must be securely hashed. Authentication should use JWT tokens. Sensitive API keys must not be hard-coded in source code and should be stored using environment variables.	Passwords are hashed, protected routes require valid authentication, and API keys are stored outside the source code.
4	Reliability & Availability	The API should handle invalid requests and server/API errors gracefully without crashing the entire application.	The system should return appropriate HTTP status codes and meaningful error messages for failed requests.
5	Usability & Accessibility	The API responses should be clear, structured, and easy for frontend applications or API testing tools such as Postman to understand and use.	All major endpoints return consistent JSON responses with understandable success and error messages.
6	Maintainability	The backend should use a modular structure with separate routes, controllers, models, middleware, configuration, and AI integration logic.	New features such as additional AI learning tools should be addable without rewriting the existing authentication and material-management modules.